

Step 1

1.1 AWS Console showing EC2 instance list and public IP address

Instance summary for i-088a1d9958b5eea0d (vpn-server-MIS420KS)

Updated less than a minute ago

Instance ID
i-088a1d9958b5eea0d

Public IPv4 address
54.242.249.168

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-22-86.ec2.internal

Instance type
t2.micro

VPC ID
vpc-0c177a7344b380a0

Subnet ID
subnet-061c4de9182cd069d

Instance ARN
arn:aws:ec2:us-east-1:655024071460:instance/i-088a1d9958b5eea0d

Private IPv4 addresses
172.31.22.86

Public IPv4 DNS
ec2-54-242-249-168.compute-1.amazonaws.com

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.

Auto Scaling Group name
-

Managed
false

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance details

AMI ID
ami-06e5a963b2dada6f

AMI name
OpenVPN Access Server Community Image-fe8020db-5343-4c43-9e65-5ed4825c931

Stop protection
Disabled

Instance auto-recovery
Default

AMI Launch index
0

Credit specification
standard

Usage operation
Shutdown instance

Monitoring
disabled

Allowed image
-

Launch time
Sat May 10 2025 16:22:20 GMT-0700 (Pacific Daylight Time) (about 23 hour s)

Lifecycle
normal

Key pair assigned at launch
vpn-keypair-MIS420KS

Kernel ID
-

RAM disk ID
-

Platform details
Linux/UNIX

Termination protection
Disabled

AMI location
aws-marketplace/OpenVPN Access Server Community Image-fe8020db-5343-4c43-9e65-5ed4825c931

Stop-hibernate behavior
Disabled

State transition reason
-

State transition message
-

Owner
655024071460

1.2 AMI selection screen showing OpenVPN Access Server from AWS Marketplace

Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Selected AMI: ami-0f88b08715681c931

Quick Start AMIs (0) | **My AMIs (0)**

Refine results

Categories
Infrastructure (80)
DevOps (33)
IoT (18)
Business Applications (33)
Cloud Operations (2)

Publisher
☐ Gigamon (27)
☐ AWS IoT Imaging (8)
☐ OpenVPN Inc. (8)
☐ HPAMI (6)
☐ Tidal Media Inc. (5)
☐ Art Group (4)
☐ Arkforce LLC (3)
☐ Hergate (2)
☐ Cloud Infrastructure Services (2)
☐ Futerale (1)
[Show All](#)

Pricing model
☐ Usage Based (70)
☐ PayPerHour (30)
☐ Bring Your Own License (2)

Pricing unit
☐ Custom Units (1)

Operating system
☐ All Linux/Unix
☐ All Windows

OpenVPN Access Server / Self-Hosted VPN (BYOL)

Overview | Product details | Pricing | Usage | Support

OpenVPN Access Server is an enterprise-grade business software VPN solution that provides a securely encrypted connection to private networks over an unsecured network such as the internet.

Typical total price
\$0.023/Hr
Total pricing per instance for services hosted on t2.xsmall in us-east-1.
[See additional pricing information.](#)

Latest version
2.13.1

Delivery methods
Amazon Machine Image (1)
Operating systems
Ubuntu 22.04.4 LTS
Ubuntu 22.04.3 LTS

Video
[Product Video](#)

Categories
Security
Network Infrastructure
Device Connectivity

Subscription notice:
A subscription to this AMI is required before you can launch an instance. Check the pricing details in the pricing tab before continuing. You can subscribe to this AMI now or we will automatically subscribe for you when you launch this instance. We recommend that you "Subscribe now" if you are sure this is the AMI you want to use to launch as it will reduce wait time on launch. Choose "Subscribe on instance launch" if you are still choosing an AMI and don't want to commit to a subscription yet. By subscribing to this AMI you agree that your use of this software is subject to the pricing terms and the seller's [End User License Agreement](#).

Buttons: Cancel | **Subscribe on instance launch** | **Subscribe now**

OpenVPN Access Server (3 Connected Devices) / Self-Hosted VPN

Free Trial
Starting from \$0.16/Hr or from \$480.00/Hr (23% savings) for software + AWS usage fees
This offering is locked to 5 connections. If you need the flexibility to scale, we recommend our PAYG listing instead.
<https://aws.amazon.com/marketplace/gp/detail/arn:aws-marketplace:us-east-1::OpenVPN:OpenVPN-Access-Server-for-AWS-delivers-the-best-of-both-worlds-VPN-solution-for-secure-remote-access,-site-to-site-VPN-and-secure-SaaS->

1.3 Detailed information of selected OpenVPN AMI image

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

🔍 Search our full catalog including 1000s of application and OS images

AMI from catalog

Recents

Quick Start

Name

OpenVPN Access Server Community Image-fe8020db-5343-4c43-9e65-5ed4a825c931

Verified provider

Description

OpenVPN Access Server 2.13.1 publisher image from <https://www.openvpn.net/>.

Image ID

ami-06e5a963b2dadea6f

Username

root

Catalog

AWS Marketplace AMIs

Published

2024-03-07T15:11:08.000Z

Architecture

x86_64

Virtualization

hvm

Root device type

ebs

ENA Enabled

Yes

Boot mode

uefi-preferred

If you have an existing license entitlement to use this software, then you can launch this software without creating a new subscription. If you do not have an existing entitlement, then by launching this software, you will be subscribed to this software and agree that your use of this software is subject to the pricing terms and the seller's [End User License Agreement](#)

🔍

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

1.4 Created key pair for EC2 instance connection

Search results

Description

OpenVPN Access Server 2.13.1 publisher image from <https://www.openvpn.net/>.

Image ID

ami-06e5a963b2dadea6f

Username

root

Catalog

AWS Marketplace AMIs

Published

2024-03-07T15:11:08.000Z

Architecture

x86_64

Virtualization

hvm

Root device type

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ENA Enabled

Yes

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▼ Instance type

Info | See advice

Instance type

12.micro

Farmly 12 1 vCPU 1 GB Memory Current generation free

The AMI vendor recommends using a 12.micro instance (or larger) for the best experience.

▼ Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have the private key file stored in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Key pair name - required

Select

▼ Network settings

Info

Network

vpc-dc177a73440380a0

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Additional charges apply when outside of free tier allowance

▼ Summary

Info

Number of instances

1

Software image (AMI)

OpenVPN Access Server / Self-Help [read more](#)

Virtual server type (Instance type)

12.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of 12.micro instance usage (or 15.micro where 12.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million LPIs, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet.

Cancel

Launch instance

Preview code

Create key pair

Key pair name

vpn-keypair-M5420K5

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

rsa RSA encrypted private and public key pair

ed25519 ED25519 encrypted private and public key pair

Private key file format

pem For use with OpenSSL

ppk For use with PuTTY

When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel Create key pair

1.5 Network settings showing public IP assignment and VPC setup

▼ Network settings Info

Edit

Network Info

vpc-0c1f77a7344b380a0

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'OpenVPN Access Server / Self-Hosted VPN (BYOL)-2.13.1-AutogenByAWSMP--1' with the following rules:

☒ Allow SSH traffic from

Recommended rule from AMI

Anywhere

0.0.0.0/0

☒ Allow CUSTOMTCP traffic from

Recommended rule from AMI

Anywhere

0.0.0.0/0

☒ Allow CUSTOMTCP traffic from

Recommended rule from AMI

Anywhere

0.0.0.0/0

☒ Allow CUSTOMUDP traffic from

Recommended rule from AMI

Anywhere

0.0.0.0/0

☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

1.6 Security group configuration with required ports for VPN and admin access

Search results

Security group name - required

OpenVPN Access Server / Self-Hosted VPN (BYOL)-2.13.1-AutogenByAWSMP--1 created 2025-05-10T12

Description - optional

OpenVPN Access Server / Self-Hosted VPN (BYOL)-2.13.1-AutogenByAWSMP--1 created 2025-05-10T12

Inbound Security Group Rules

Security group rule 1 (TCP) 22, 68, 100, 79, 131/32

Type

SSH

Protocol

TCP

Port range

22

Source type

My IP

Name

68, 100, 79, 131/32

Description - optional

e.g. SSH for admin desktop

Security group rule 2 (TCP) 443, 0.0.0.0/0

Type

HTTPS

Protocol

TCP

Port range

443

Source type

Anywhere

Name

0.0.0.0/0

Description - optional

e.g. SSH for admin desktop

Security group rule 3 (UDP) 1194, 0.0.0.0/0

Type

Custom UDP

Protocol

UDP

Port range

1194

Source type

Anywhere

Name

0.0.0.0/0

Description - optional

e.g. SSH for admin desktop

Security group rule 4 (TCP) 943, 68, 100, 79, 131/32

Type

Custom TCP

Protocol

TCP

Port range

943

Source type

My IP

Name

68, 100, 79, 131/32

Description - optional

e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

Summary

Number of instances

1

Software Image (AMI)

OpenVPN Access Server / Self-Hosted VPN (BYOL)-2.13.1-AutogenByAWSMP--1 created 2025-05-10T12

Virtual server type (Instance type)

t2.micro

Firewall (Security group)

New security group

Storage (EBS volume)

1 volume(s) - 8 GB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or 15 minutes where t2.micro isn't available) when used with Free tier AMIs, 100 hours per month of public IP address usage, 50 GB of EBS storage, 2 million IOPS, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Launch instance

Preview code

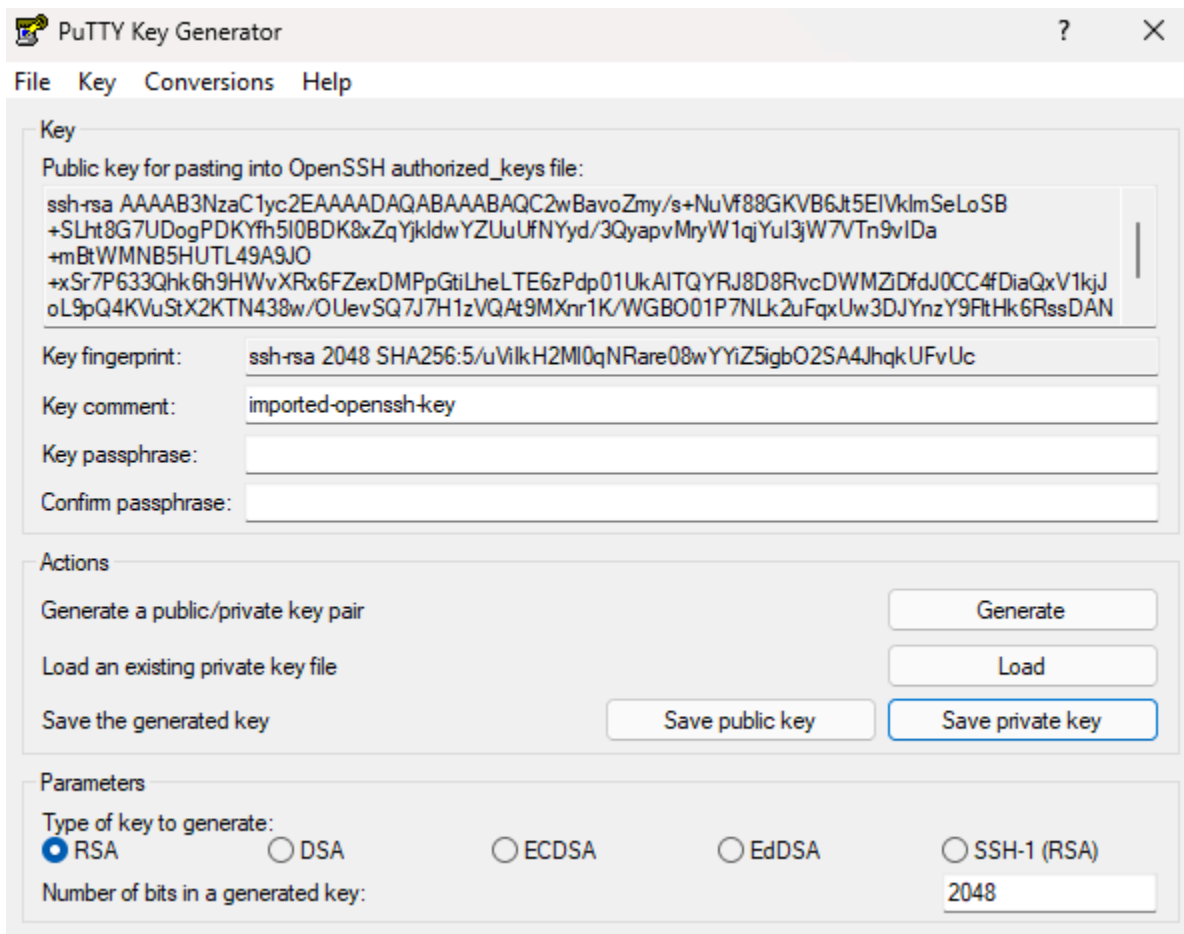
1.7 EC2 instance running with instance ID, IP, and security group visible



Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name	Launch time
vpn-keppai-M1	i-08ba1d958b0cead0f	Running	t2.micro		View alarms	us-east-1d	ec2-54-242-249-168.co...	54-242-249-168	-	-	disabled	vpn-ig-MIS42DKS	vpn-keppai-M1...	2025/05/10 16:22 GMT-7

Step 2

2.1 Successful SSH login into EC2 instance as openvpnas user



PuTTY Key Generator

File Key Conversions Help

Key

Public key for pasting into OpenSSH authorized_keys file:

```
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQAC2wBavoZmy/s+NuVf88GKVB6Jt5EIVklmSeLoSB
+SLht8G7UDogPDKYfh5I0BDK8xZqYjklwYZUuUfNYyd/3QyapvMryW1qYul3jW7Vtn9vIDa
+mBtWMNB5HUTL49A9JO
+XSr7P633Qhk6h9HWvXR6FZexDMPpGtiLheLTE6zPdp01UkAITQYRJ8D8RvcDWMZiDfdJ0CC4fDiaQxV1kjJ
oL9pQ4KVuStX2KTN438w/OUevSQ7J7H1zVQAt9MXnr1K/WGBO01P7NLk2uFqxUw3DJYnzY9FItHk6RssDAN
```

Key fingerprint: ssh-rsa 2048 SHA256:5/uVilkH2MI0qNRare08wYYiZ5igbO2SA4JhqkUFvUc

Key comment: imported-openssh-key

Key passphrase:

Confirm passphrase:

Actions

Generate a public/private key pair Generate

Load an existing private key file Load

Save the generated key Save public key Save private key

Parameters

Type of key to generate:

☒ RSA ☐ DSA ☐ ECDSA ☐ EdDSA ☐ SSH-1 (RSA)

Number of bits in a generated key: 2048

```
openvpnas@ip-172-31-22-86: ~  
/lib/systemd/system/openvpnas.service.  
Starting openvpnas...  
  
NOTE: Your system clock must be correct for OpenVPN Access Server  
to perform correctly. Please ensure that your time and date  
are correct on this system.  
  
Initial Configuration Complete!  
  
You can now continue configuring OpenVPN Access Server by  
directing your Web browser to this URL:  
  
https://54.242.249.168:943/admin  
  
During normal operation, OpenVPN AS can be accessed via these URLs:  
Admin UI: https://54.242.249.168:943/admin  
Client UI: https://54.242.249.168:943/  
To login please use the "openvpn" account with "KkV33wUCBr5S" password.  
  
See the Release Notes for this release at:  
https://openvpn.net/vpn-server-resources/release-notes/  
  
openvpnas@ip-172-31-22-86:~$ ^C  
openvpnas@ip-172-31-22-86:~$
```

2.2 Terminal output showing OpenVPN admin password after initial setup

```
openvpnas@ip-172-31-22-86: ~  
/lib/systemd/system/openvpnas.service.  
Starting openvpnas...  
  
NOTE: Your system clock must be correct for OpenVPN Access Server  
to perform correctly. Please ensure that your time and date  
are correct on this system.  
  
Initial Configuration Complete!  
  
You can now continue configuring OpenVPN Access Server by  
directing your Web browser to this URL:  
  
https://54.242.249.168:943/admin  
  
During normal operation, OpenVPN AS can be accessed via these URLs:  
Admin UI: https://54.242.249.168:943/admin  
Client UI: https://54.242.249.168:943/  
To login please use the "openvpn" account with "KkV33wUCBr5S" password.  
  
See the Release Notes for this release at:  
https://openvpn.net/vpn-server-resources/release-notes/
```

2.3 OpenVPN admin panel login screen

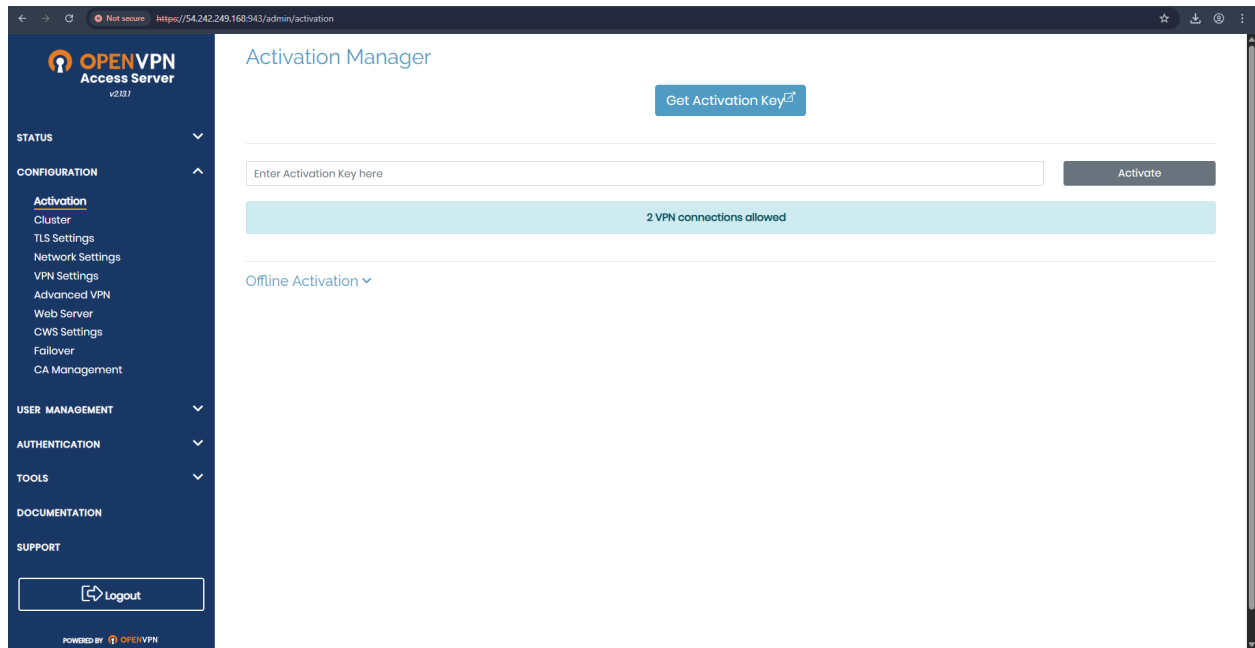
```
openvpnas@ip-172-31-22-86: ~  
/lib/systemd/system/openvpnas.service.  
Starting openvpnas...  
  
NOTE: Your system clock must be correct for OpenVPN Access Server  
to perform correctly. Please ensure that your time and date  
are correct on this system.  
  
Initial Configuration Complete!  
  
You can now continue configuring OpenVPN Access Server by  
directing your Web browser to this URL:  
  
https://54.242.249.168:943/admin  
  
During normal operation, OpenVPN AS can be accessed via these URLs:  
Admin UI: https://54.242.249.168:943/admin  
Client UI: https://54.242.249.168:943/  
To login please use the "openvpn" account with "KkV33wUCBr5S" password.  
  
See the Release Notes for this release at:  
https://openvpn.net/vpn-server-resources/release-notes/
```



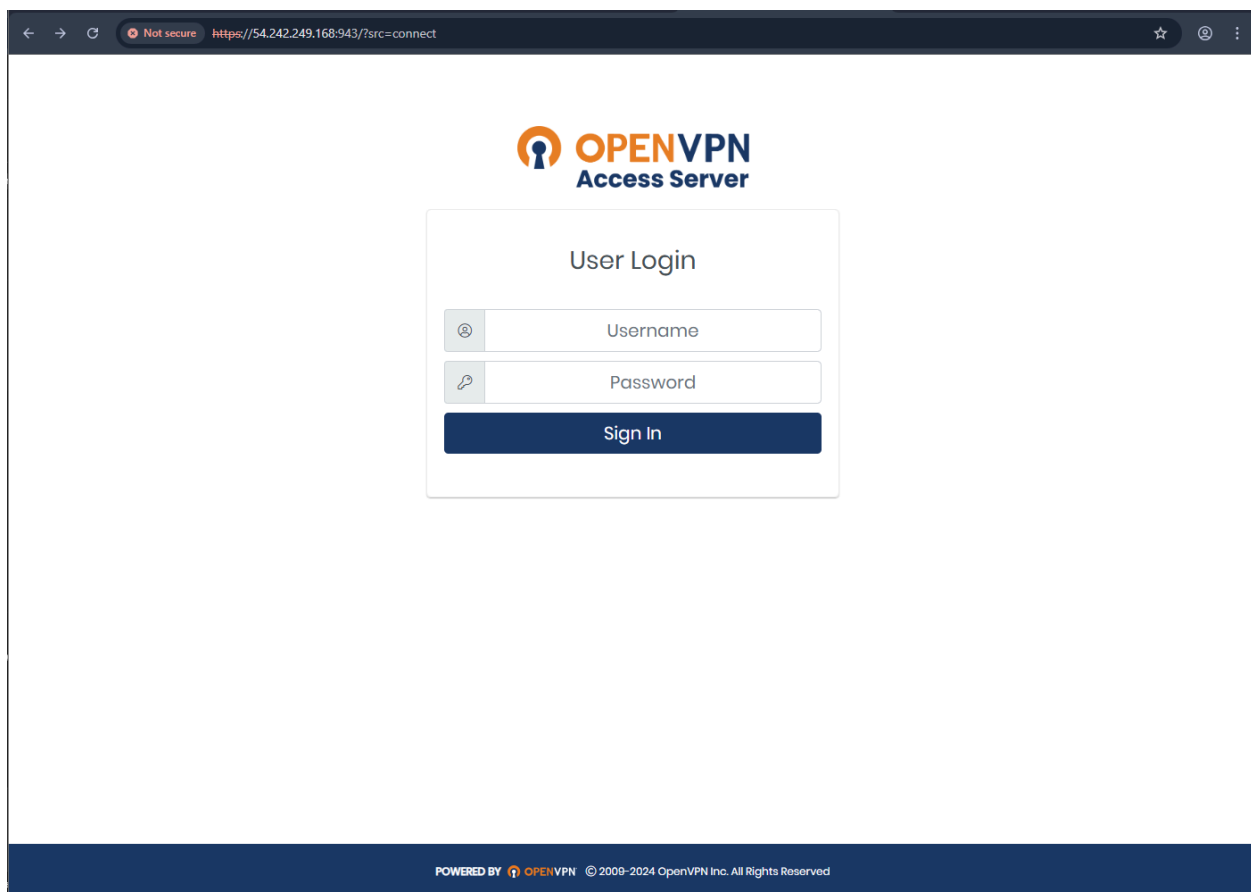
Admin Login

Incorrect login.

Sign In

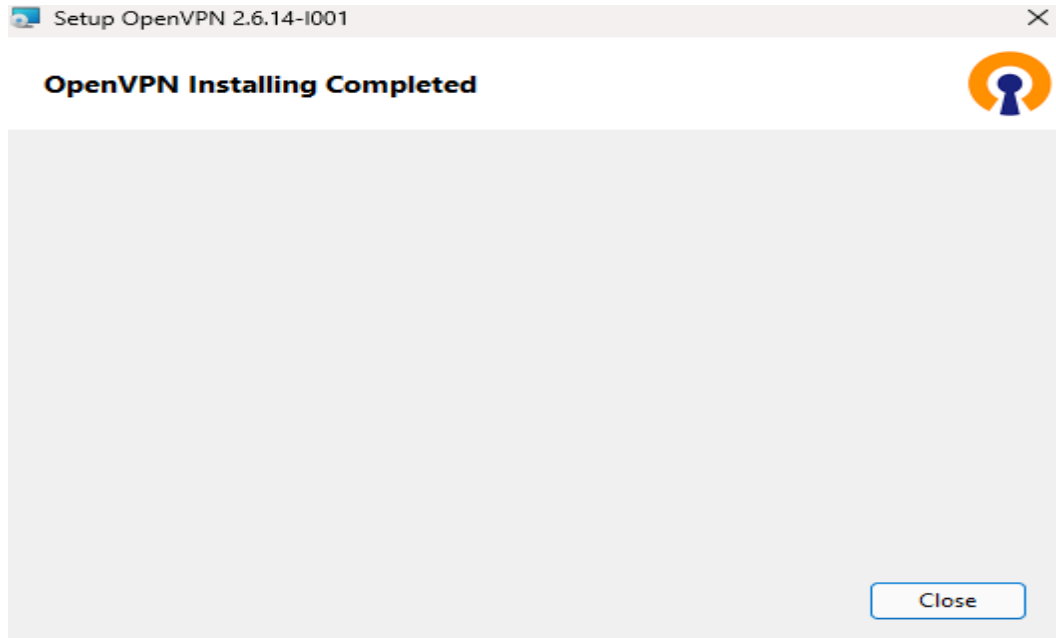


2.4 OpenVPN client panel for downloading connection profiles

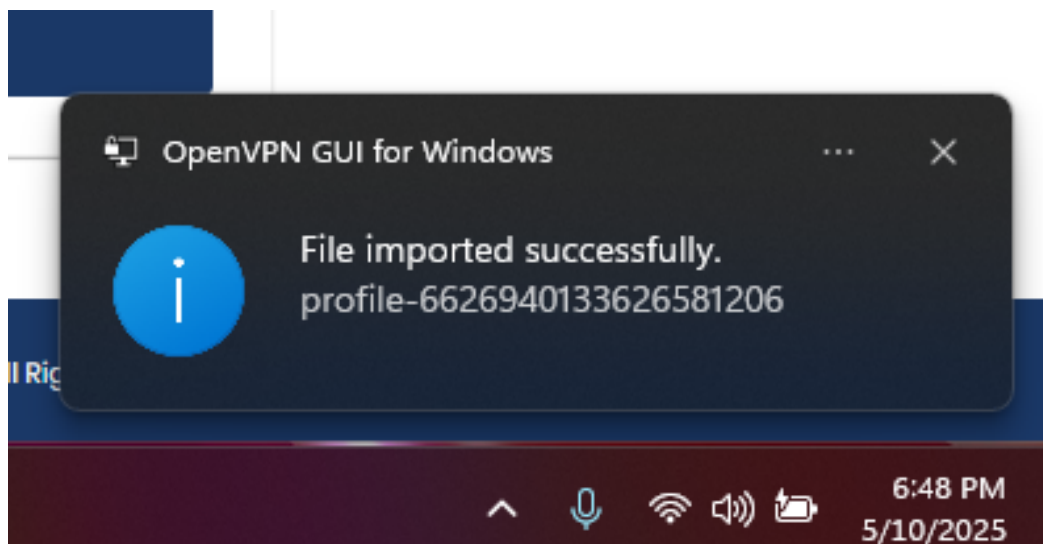


Step 3

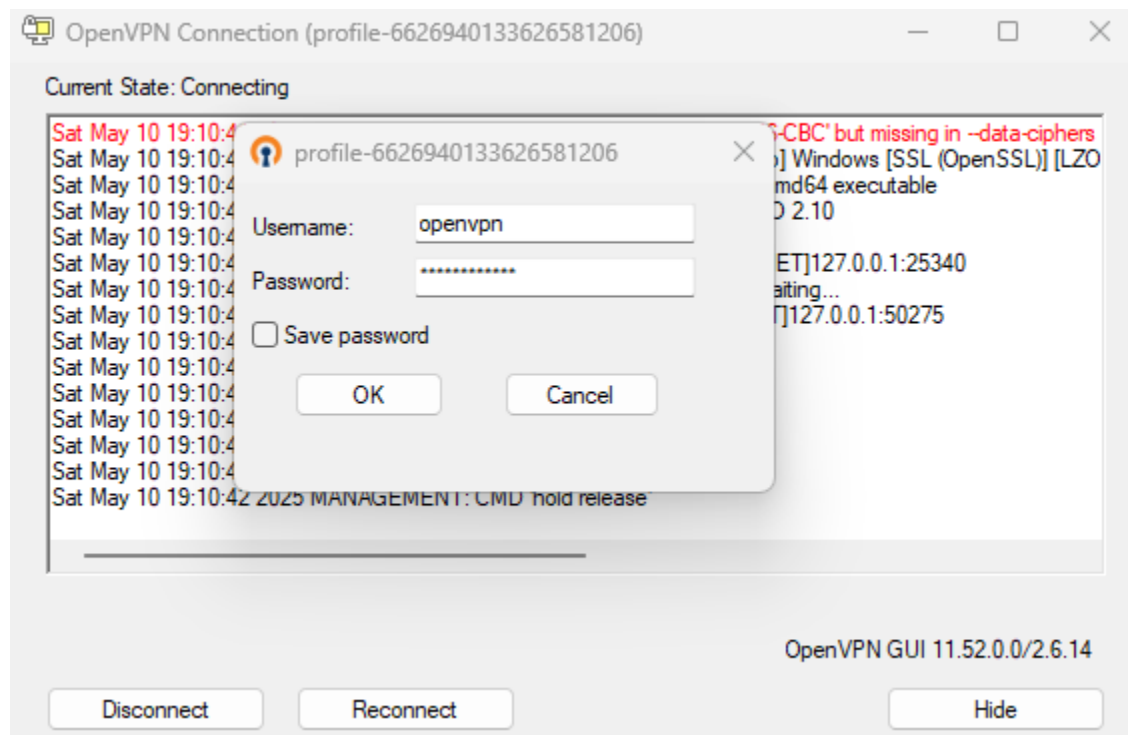
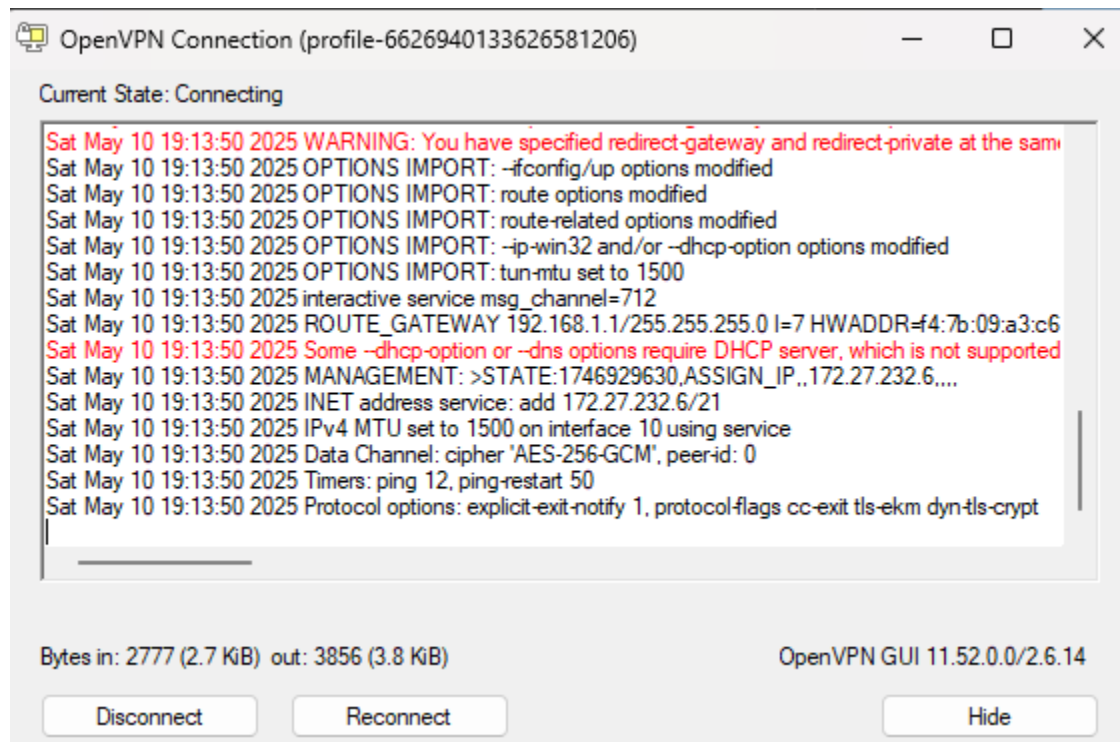
3.1 Confirmation of OpenVPN client installed on system



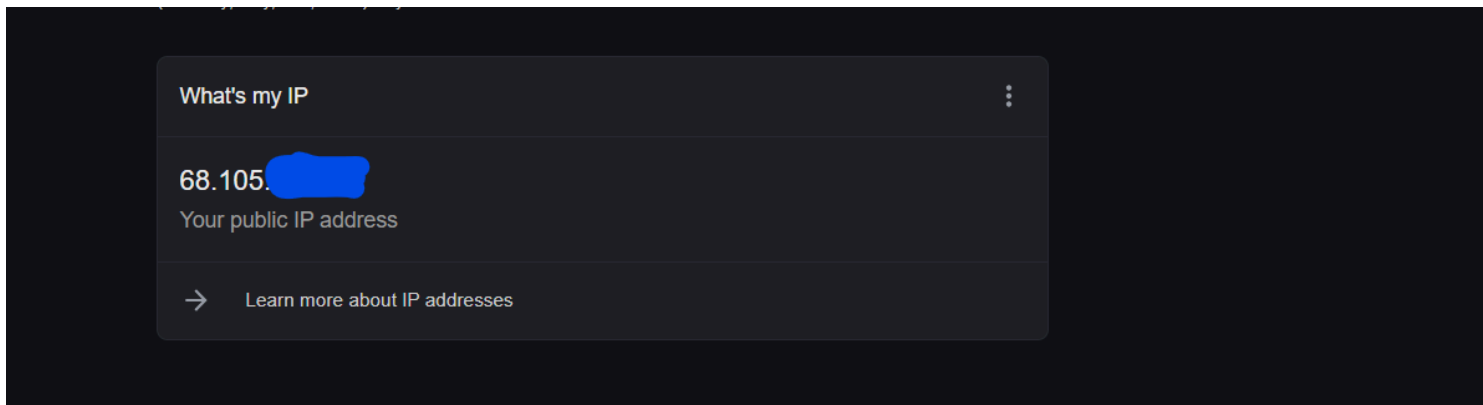
3.2 OpenVPN profile imported successfully into OpenVPN Connect



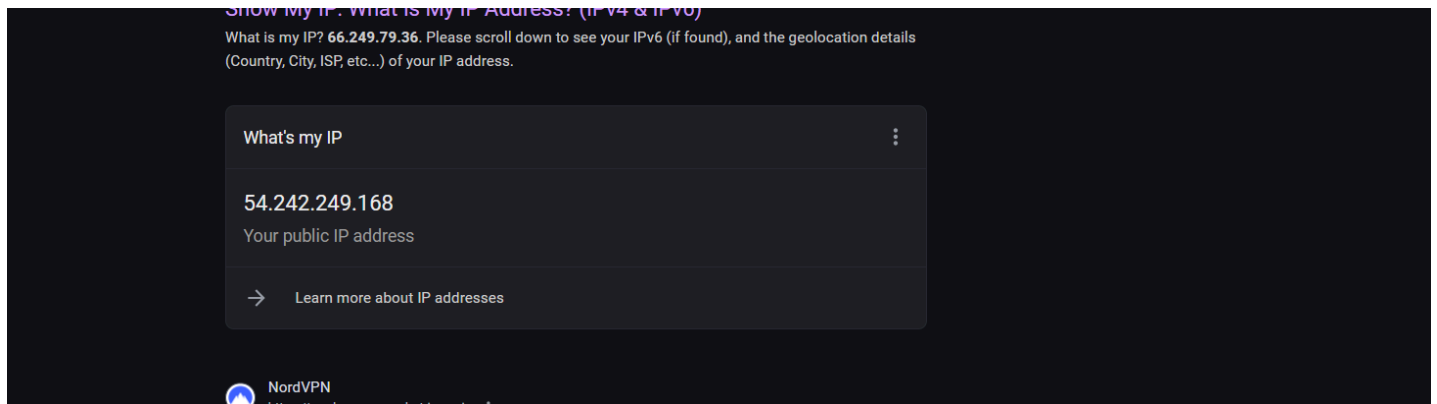
3.3 OpenVPN connection established client connected



3.4 Computer IP address before connecting to VPN



3.5 Computer IP address after connecting to VPN where it masked my Ip address



Step 4

4.1 S3 bucket created for VPN backups

aws

Search

[Alt+S]

United States (N. Virginia)

Karlos

Amazon S3

Buckets

Successfully created bucket "vpn-backups-mis420ks"

To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Account snapshot - updated every 24 hours

All AWS Regions

[View Storage Lens dashboard](#)

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets

Directory buckets

General purpose buckets (1)

Info

All AWS Regions

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

Name	AWS Region	IAM Access Analyzer	Creation date
vpn-backups-mis420ks	US East (N. Virginia) us-east-1	View analyzer for us-east-1	May 10, 2025, 20:09:46 (UTC-07:00)

4.2 S3 bucket created for VPN logs

aws

Search

[Alt+S]

United States (N. Virginia)

Karlos

Amazon S3

Buckets

Successfully created bucket "vpn-logs-mis420ks"

To upload files and folders, or to configure additional bucket settings, choose [View details](#).

View details

Account snapshot - updated every 24 hours

All AWS Regions

View Storage Lens dashboard

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets

Directory buckets

General purpose buckets (2)

Info

All AWS Regions

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

	Name	AWS Region	IAM Access Analyzer	Creation date
☐	vpn-backups-mis420ks	US East (N. Virginia) us-east-1	View analyzer for us-east-1	May 10, 2025, 20:09:46 (UTC-07:00)
☐	vpn-logs-mis420ks	US East (N. Virginia) us-east-1	View analyzer for us-east-1	May 10, 2025, 20:12:28 (UTC-07:00)

4.3 Server-side encryption enabled on backups bucket

Amazon S3 > Buckets > vpn-backups-mis420ks > Edit default encryption

1

2

3

Edit default encryption

Info

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Info

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

☒ Enable

Cancel

Save changes

4.4 Server side encryption enabled on logs bucket

aws

Search

[Alt+S]

United States (N. Virginia)

Karlos

Amazon S3 > Buckets > vpn-logs-mis420ks > Edit default encryption

1

2

3

Edit default encryption

Info

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Info

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

☒ Enable

Cancel

Save changes

Step 5

5.1 IAM user created for admin access vpn-admin

[Alt+S]

United States (N. Virginia)

Karlos

vpn-backups-mis420ks

vpn-backups-mis420ks

ObjectsMetadataPropertiesPermissionsMetricsManagementAccess Points

Bucket overview

AWS Region

US East (N. Virginia) us-east-1

Amazon Resource Name (ARN)

arn:aws:s3::vpn-backups-mis420ks

Creation date

May 10, 2025, 20:09:46 (UTC-07:00)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

Enabled

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Disabled

Tags (0)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

Key

Value

No tags associated with this resource.

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Server-side encryption with Amazon S3 managed keys (SSE-S3)

Bucket Key

When KMS encryption is used to encrypt new objects in this bucket, the bucket key reduces encryption costs by lowering calls to AWS KMS. [Learn more](#)

Enabled

Intelligent-Tiering Archive configurations (0)

View details

Edit

Delete

Create configuration

Enable objects stored in the Intelligent-Tiering storage class to tier-down to the Archive Access tier or the Deep Archive Access tier which are optimized for objects that will be rarely accessed for long periods of time. [Learn more](#)

Find Intelligent-Tiering Archive configurations

Name	Status	Scope	Days until transition to Archive Acces...	Days until transition to Deep Archive...
------	--------	-------	---	--

5.2 IAM user created for log viewing vpn logviewer

The screenshot shows the AWS IAM console interface for the user 'vpn-logviewer'. The left sidebar contains navigation links for Access, Groups, Tags, Security credentials, and Last Accessed. The main content area displays the user's summary, permissions, and options to generate a policy based on CloudTrail events.

Summary

- ARN: `arn:aws:iam::655024071460:user/vpn-logviewer`
- Console access: Disabled
- Access key 1: [Create access key](#)
- Created: May 10, 2025, 21:33 (UTC-07:00)
- Last console sign-in: -

Permissions

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

Policy name	Type	Attached via
AmazonS3ReadOnlyAccess	AWS managed	Directly

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

[Generate policy](#)

No requests to generate a policy in the past 7 days.

5.3 IAM role created with S3 full access VPNAdminRole

The screenshot shows the AWS IAM console interface for the role 'vpn-admin'. The left sidebar contains navigation links for Access, Groups, Tags, Security credentials, and Last Accessed. The main content area displays the role's summary, permissions, and options to generate a policy based on CloudTrail events.

Summary

- ARN: `arn:aws:iam::655024071460:role/vpn-admin`
- Console access: Disabled
- Access key 1: [Create access key](#)
- Created: May 10, 2025, 21:31 (UTC-07:00)
- Last console sign-in: -

Permissions

Permissions policies (2)

Permissions are defined by policies attached to the role directly or through groups.

Filter by Type: All types

Policy name	Type	Attached via
AmazonEC2ReadOnlyAccess	AWS managed	Directly
AmazonS3FullAccess	AWS managed	Directly

Permissions boundary (not set)

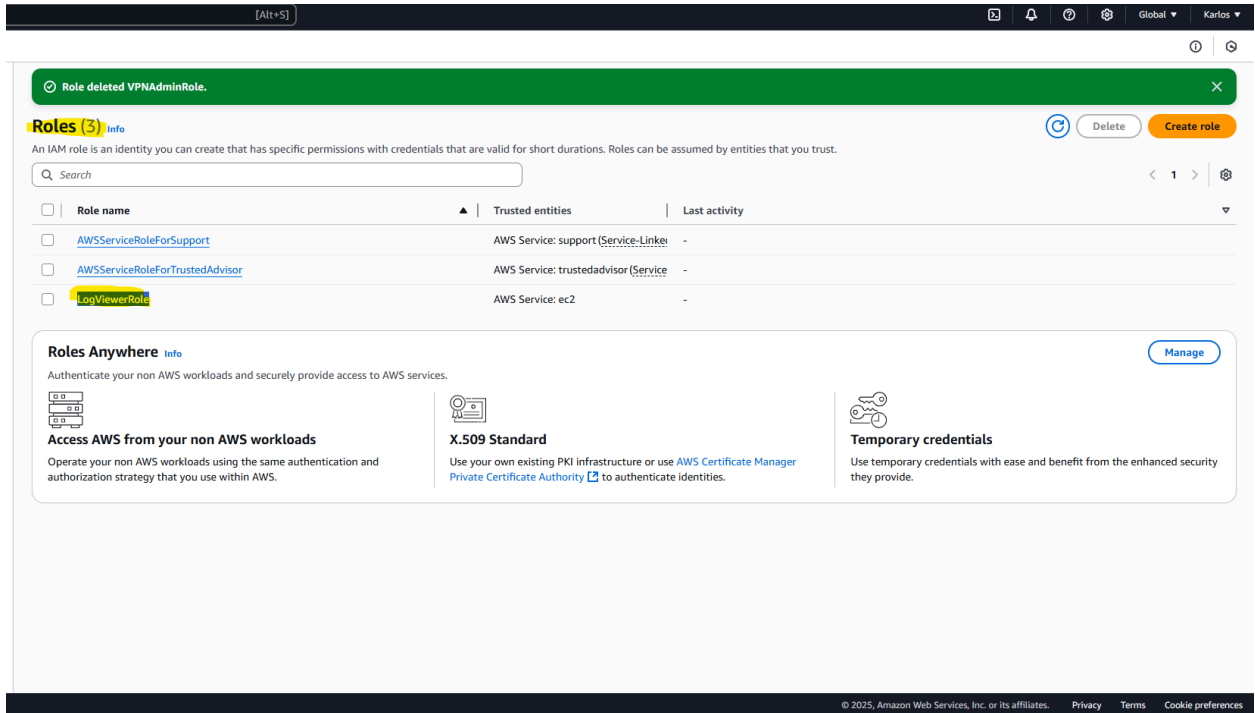
Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

[Generate policy](#)

No requests to generate a policy in the past 7 days.

5.4 IAM role created for log viewing permissions LogViewerRole



Role deleted VPNAdminRole.

Roles (3) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linker)	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linker)	-
☐	LogViewerRole	AWS Service: ec2	-

Roles Anywhere [Info](#) [Manage](#)

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.

X.509 Standard

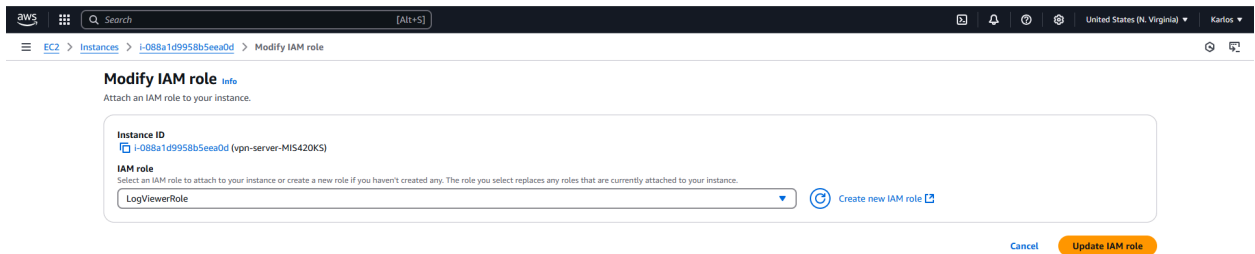
Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.

Temporary credentials

Use temporary credentials with ease and benefit from the enhanced security they provide.

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5.5 EC2 instance role successfully updated to include IAM role



Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID
[i-088a1d9958b5eea0d](#) (vpn-server-MIS420KS)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

[LogViewerRole](#) [Create new IAM role](#)

[Cancel](#) [Update IAM role](#)

Alt+S

GlobalKarlos

VPNAdminRole

Delete

Allows EC2 instances to call AWS services on your behalf.

Summary

Creation date

May 11, 2025, 17:09 (UTC-07:00)

ARN

arn:aws:iam::655024071460:role/VPNAdminRole

Instance profile ARN

arn:aws:iam::655024071460:instance-profile/VPNAdminRole

Last activity

-

Maximum session duration

1 hour

Edit

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (2)

info

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type

All types

< 1 >

☐

Policy name

Type

Attached entities

☐

AmazonInspector2ReadOnlyAccess

AWS managed

1

☐

AmazonS3FullAccess

AWS managed

2

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

5.6 IAM role created with S3 read-only access VPNUserRole

Alt+S

GlobalKarlos

Role VPNUserRole created.

View role

X

VPNUserRole

info

Delete

IAM role created to simulate a least-privilege user, with S3 read-only permissions for minimal access scenarios such as log viewing or backup status.

Summary

Creation date

May 11, 2025, 17:12 (UTC-07:00)

ARN

arn:aws:iam::655024071460:role/VPNUserRole

Instance profile ARN

arn:aws:iam::655024071460:instance-profile/VPNUserRole

Last activity

-

Maximum session duration

1 hour

Edit

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1)

info

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type

All types

< 1 >

☐

Policy name

Type

Attached entities

☐

AmazonS3ReadOnlyAccess

AWS managed

3

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

Step 6

6.1 CloudWatch log group created for collecting logs from EC2

The screenshot shows the 'Add metric and conditions' step of the 'Create alarm' wizard in AWS CloudWatch. The 'Metric' section on the left displays a line graph for 'CPUUtilization' with a red threshold line at 5%. The 'Conditions' section on the right is configured with a 'Static' threshold type, 'Greater' comparison, and a threshold value of 5. The 'Additional configuration' section is expanded, showing 'Namespace' as 'AWS/EC2', 'Metric name' as 'CPUUtilization', 'Instanceid' as 'i-088a1d9958b5eca0d', and 'Instance name' as 'vpn-server-mis420ks'. The 'Statistic' is set to 'Average' and the 'Period' is '5 minutes'.

Metric

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent

7.85

4.98

2.11

07:30 08:00 08:30 09:00 09:30 10:00 10:30

CPUUtilization

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...

Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
>= threshold

☐ Lower/Equal
<= threshold

☐ Lower
< threshold

than...

Define the threshold value.

5

Must be a number

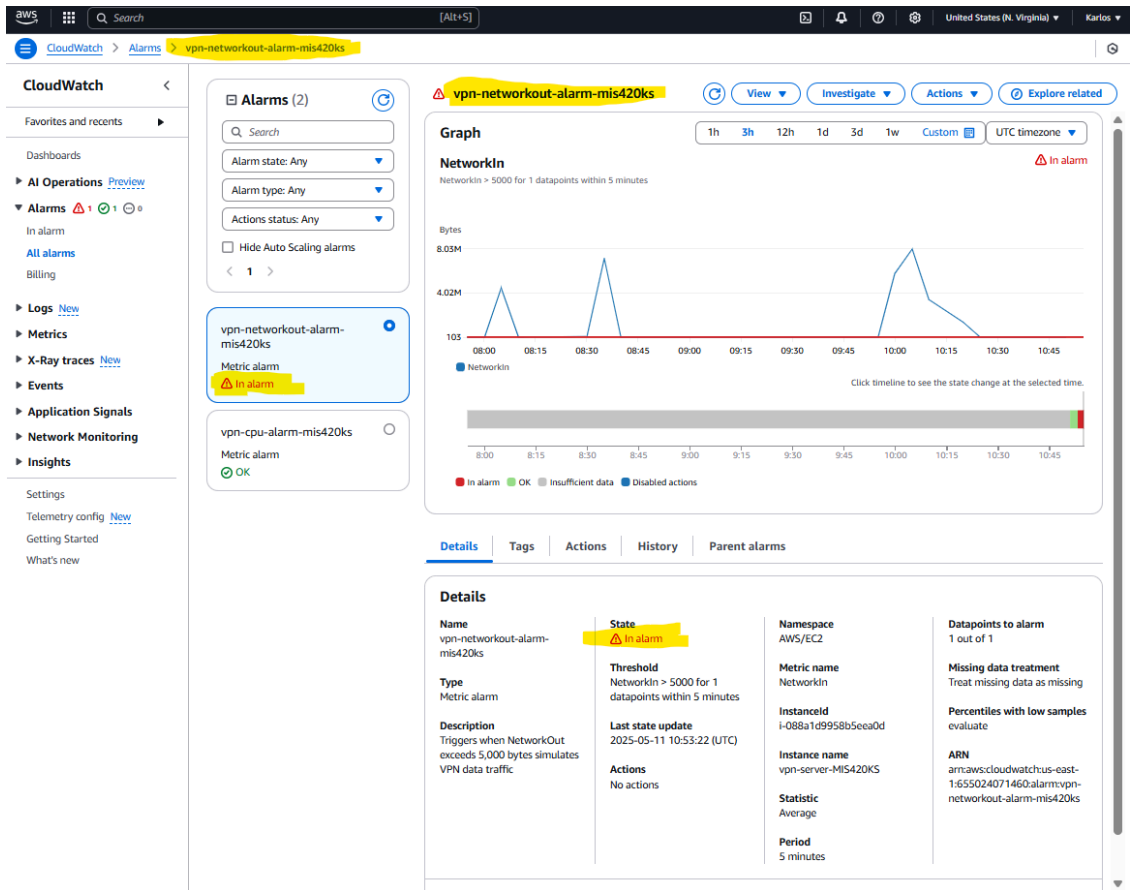
► Additional configuration

Cancel Next

6.2 CloudWatch alarm configured for CPU usage to simulate login attempts

The screenshot shows the 'Log groups' page in the AWS CloudWatch console. A green notification banner at the top states 'Log group "/>

6.3 CloudWatch alarm created for bandwidth/network traffic monitoring

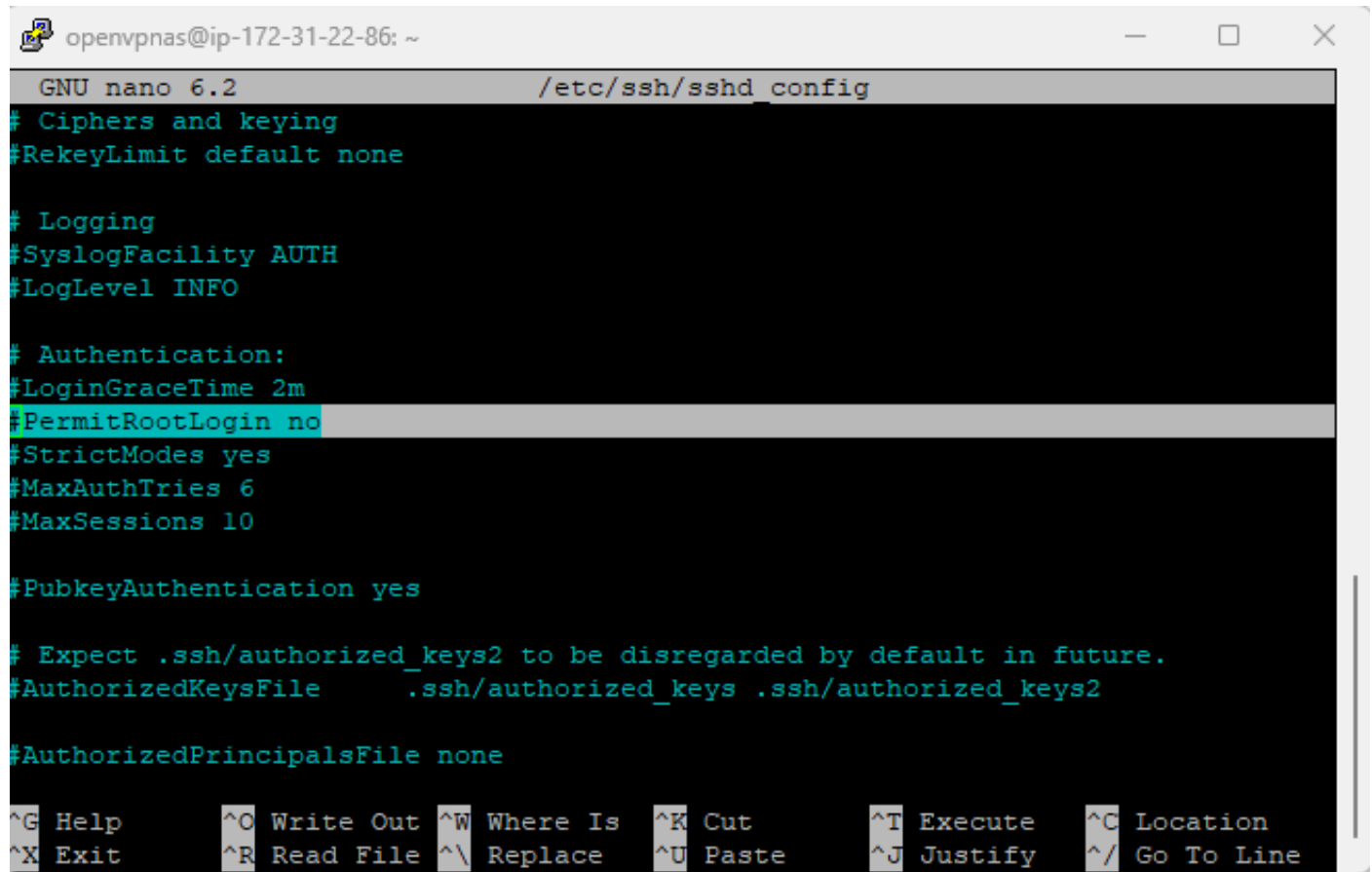


7.1 AWS Inspector scan results showing open ports and network risks

7.1 AWS Inspector scan results showing open ports and network risks

Step 8

8.1 Root login disabled in SSH configuration file



The screenshot shows a terminal window with the title bar "openvpn@ip-172-31-22-86: ~". The window contains the SSH configuration file `/etc/ssh/sshd_config` being edited with GNU nano 6.2. The configuration is as follows:

```
GNU nano 6.2 /etc/ssh/sshd_config
# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:
#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

The line `#PermitRootLogin no` is highlighted in blue. The terminal window has standard window controls (minimize, maximize, close) in the top right corner.

8.2 Password authentication disabled for SSH access

```
openvpnas@ip-172-31-22-86: ~
openvpnas@ip-172-31-22-86:~$ sudo systemctl restart ssh
openvpnas@ip-172-31-22-86:~$ sudo nano /etc/ssh/sshd_config
openvpnas@ip-172-31-22-86:~$ grep "^PermitRootLogin" /etc/ssh/sshd_config
openvpnas@ip-172-31-22-86:~$ grep "^PermitRootLogin" /etc/ssh/sshd_config
openvpnas@ip-172-31-22-86:~$
openvpnas@ip-172-31-22-86:~$
openvpnas@ip-172-31-22-86:~$
openvpnas@ip-172-31-22-86:~$
openvpnas@ip-172-31-22-86:~$ grep "^PermitRootLogin" /etc/ssh/sshd_config
openvpnas@ip-172-31-22-86:~$ sudo nano /etc/ssh/sshd_config
openvpnas@ip-172-31-22-86:~$ sudo nano /etc/ssh/sshd_config
openvpnas@ip-172-31-22-86:~$ sudo systemctl restart ssh
openvpnas@ip-172-31-22-86:~$ grep "^PermitRootLogin" /etc/ssh/sshd_config
PermitRootLogin no
openvpnas@ip-172-31-22-86:~$ ^C
openvpnas@ip-172-31-22-86:~$ ^C
openvpnas@ip-172-31-22-86:~$ ^C
openvpnas@ip-172-31-22-86:~$ sudo nano /etc/ssh/sshd_config
openvpnas@ip-172-31-22-86:~$ sudo systemctl restart ssh
openvpnas@ip-172-31-22-86:~$ grep "^PasswordAuthentication" /etc/ssh/sshd_config

PasswordAuthentication no
openvpnas@ip-172-31-22-86:~$
```

8.3 Firewall rules configured using UFW to allow required VPN and SSH ports

```
openvpnas@ip-172-31-22-86: ~  
22/tcp ALLOW Anywhere  
443/tcp ALLOW Anywhere  
943/tcp ALLOW Anywhere  
1194/udp ALLOW Anywhere  
22/tcp (v6) ALLOW Anywhere (v6)  
443/tcp (v6) ALLOW Anywhere (v6)  
943/tcp (v6) ALLOW Anywhere (v6)  
1194/udp (v6) ALLOW Anywhere (v6)  
  
openvpnas@ip-172-31-22-86:~$ sudo ufw status  
Status: active  
  
To Action From  
--  
22/tcp ALLOW Anywhere  
443/tcp ALLOW Anywhere  
943/tcp ALLOW Anywhere  
1194/udp ALLOW Anywhere  
22/tcp (v6) ALLOW Anywhere (v6)  
443/tcp (v6) ALLOW Anywhere (v6)  
943/tcp (v6) ALLOW Anywhere (v6)  
1194/udp (v6) ALLOW Anywhere (v6)  
  
openvpnas@ip-172-31-22-86:~$
```

Step 9

9.1 SNS email subscription confirmations for CloudWatch alerts

